CHAPTER 12

KINETIC THEORY

CHAPTER ESSENCE AT A GLANCE:
Gases are billions of microscopic billiard balls bouncing endlessly! Pressure is simply the impact of countless particles slamming against the container walls. Temperature is nothing more than the average kinetic energy of these zooming particles!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Gas Pressure Formula:** P = ρ v rms^2 = (M divided by V) v rms^2. Pressure originates from molecular collisions against container walls.

• **Temperature Interpretation:** Average translational kinetic energy per molecule: E = ^{3} divided by $_2$ k B T. Absolute temperature T is a direct measure of average molecular kinetic energy!

• **RMS Speed & Effusion:** v rms = sqrt(3 R T divided by M) proportional to 1 divided by sqrt(M). Since Hydrogen (M=2) is 16 times lighter than Oxygen (M=32), hydrogen molecules move 4 times faster and leak through pinholes 4× faster!

• **Degrees of Freedom & γ:** Monatomic gas (He, Ar): f = 3, C v = ^{3} divided by $_2$ R, C p = ^{5} divided by $_2$ R, γ = 1.67. Rigid Diatomic gas (O_2, N_2): f = 5 (3 trans + 2 rot), C v = ^{5} divided by $_2$ R, C p = ^{7} divided by $_2$ R, γ = 1.40.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$P = \frac{1}{3}\rho v_{\text{rms}}^2 = \frac{1}{3}\frac{M}{V}v_{\text{rms}}^2$$

Kinetic Pressure of Ideal Gas

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3k_BT}{m}}$$

Root-Mean-Square Speed of Gas Molecules

$$E_{\text{avg}} = \frac{3}{2}k_BT \quad \text{per molecule}$$

Average Translational Kinetic Energy

$$\lambda = \frac{1}{\sqrt{2}\pi n d^2}$$

Mean Free Path of Gas Molecules

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** RMS speed depends on the absolute temperature T and molar mass M: v rms proportional to sqrt(T divided by M). Lighter gas molecules (like Hydrogen) fly much faster than heavy ones (like Oxygen)!

• **Alert:** At Absolute Zero (0 Kelvin), all molecular motion ceases and kinetic pressure becomes zero!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the fundamental postulates of the Kinetic Theory of an Ideal Gas. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Postulates]** 1. A gas consists of large numbers of identical, tiny, hard spherical particles called molecules in continuous random rapid motion. 2. Volume of gas molecules is negligibly small compared to volume of container. 3. Molecules exert no intermolecular forces of attraction except during collisions. 4. Collisions between molecules and with walls are perfectly elastic. [2 Marks]

• **Key Exam Result:** Random motion, negligible volume, no attractive forces, elastic collisions

2. Define Degrees of Freedom (f) of a molecule. State degrees of freedom for: (a) Monoatomic gas (b) Rigid diatomic gas. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition]** Degrees of freedom is the total number of independent coordinates or independent squared terms in the energy expression required to specify the motion and position of a molecule. [1 Mark]

• **[Step 2: Monoatomic & Diatomic]** (a) Monoatomic gas (He, Ar): f = 3 (3 translational). (b) Rigid diatomic gas (O2, N2 at room temp): f = 5 (3 translational + 2 rotational). [1 Mark]

• **Key Exam Result:** Monoatomic: f = 3; Diatomic: f = 5

3. Show that the average translational kinetic energy of an ideal gas molecule is directly proportional to absolute temperature: E = 3 divided by 2 k B T. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Kinetic Pressure Formula]** Pressure exerted by gas: P = one-third ρ v rms^2 = one-third (M divided by V) v rms^2 $\Rightarrow$ P V = one-third M v rms^2. [1 Mark]

• **[Step 2: Ideal Gas Equation]** For 1 mole of gas: P V = R T $\Rightarrow$ one-third M v rms^2 = R T $\Rightarrow$ $\frac{1}{2}$ M v rms^2 = 3 divided by 2 R T. [1 Mark]

• **[Step 3: Energy per Molecule]** Dividing by Avogadro's number N A: E = $\frac{1}{2}$ m v rms^2 = (3 divided by 2) (R divided by N A) T = 3 divided by 2 k B T. This proves temperature is a direct measure of average molecular kinetic energy. [1 Mark]

• **Key Exam Result:** E = 3 divided by 2 k B T (Translational KE is proportional to T)

4. Why does the temperature of a gas decrease during an adiabatic expansion? Explain from the viewpoint of kinetic theory and thermodynamics. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Thermodynamic View]** In adiabatic expansion, $\Delta Q = 0$. By First Law: $W = -\Delta U$. Work is done by the gas entirely at the expense of its internal energy, causing internal energy to decrease ($\Delta U < 0$). [1.5 Marks]

• **[Step 2: Kinetic Theory View]** Molecules colliding with a receding container piston bounce back with reduced speed, losing kinetic energy. Since temperature is directly proportional to average molecular kinetic energy, the reduction in molecular speed results in a drop in temperature. [1.5 Marks]

• **Key Exam Result:** Molecules lose KE doing expansion work; T falls because T proportional to KE

5. Using the Law of Equipartition of Energy, calculate the ratio of specific heats γ = C p divided by C v for: (a) A monoatomic gas (b) A rigid diatomic gas. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Monoatomic Gas (f = 3)]** Internal energy U = 3 divided by 2 R T. C v = dU divided by dT = 3 divided by 2 R. C p = C v + R = 5 divided by 2 R. Ratio γ = C p divided by C v = (5 divided by 2 R) divided by (3 divided by 2 R) = 5 divided by 3 approx. 1.67. [1.5 Marks]

• **[Step 2: Diatomic Gas (f = 5)]** Internal energy U = 5 divided by 2 R T. C v = dU divided by dT = 5 divided by 2 R. C p = C v + R = 7 divided by 2 R. Ratio γ = C p divided by C v = (7 divided by 2 R) divided by (5 divided by 2 R) = 7 divided by 5 = 1.40. [1.5 Marks]

• **Key Exam Result:** Monoatomic: γ = 5 divided by 3 = 1.67; Diatomic: γ = 7 divided by 5 = 1.40

6. Calculate the root-mean-square (rms) speed of oxygen molecules at 27 °C. (Molar mass of O2 = 32 g/mol, R = 8.31 J mol^{-1} K^{-1}) **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Given Data in SI Units]** Temperature T = 27 + 273 = 300 K. Molar mass M = 32 g/mol = 32 × 10^{-3} kg/mol. R = 8.31 J mol^{-1} K^{-1}. [1 Mark]

• **[Step 2: Formula & Substitution]** v rms = √(3 R T divided by M) = √[(3 × 8.31 × 300) divided by (32 × 10^{-3})] = √[7479 divided by (0.032)] = √(233,718.75). [1.5 Marks]

• **[Step 3: Calculation]** v rms = 483.4 m/s. [0.5 Mark]

• **Key Exam Result:** v rms = 483.4 m/s